Gas Laws Practicals

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1. Practical 1: Boyle's Law

1.1. Aim

To Investigate Boyle's Law

1.2. Introduction

Given measuring the cross sectional area of the tube is rather difficult; the area stays constant throughout, therefore the length of the fluid in the tube is proportional to the volume of the fluid in the capillary tube.

$$\therefore L \propto V$$

1.3. Variables

- **Independent:** Volume (Length)
- **Dependent:** Pressure
- **Control:** Mass, Temperature

1.4. Method

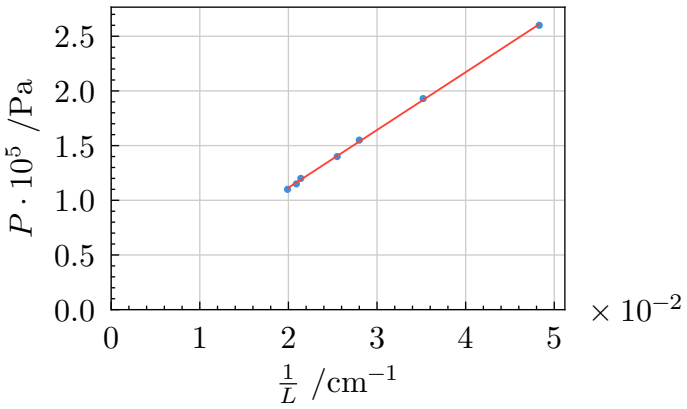
1. Set up the apparatus shown in the diagram, ensuring the tube is empty
2. Drawing in air each time to the tube (slowly), record the height (L_n /cm), until the total volume of the tube is filled up
3. Repeat the experiment removing the water in the tube and record the mean volume for each mass ($\overline{L_n}$ /cm)
4. Plot a graph of P /Pa (y-axis) against $\frac{1}{L}$ /cm^{-1} (x-axis), drawing a best-fit line, and deduce from the straight line through the origin that $P \propto \frac{1}{L}$.

1.5. Risk Assessment

Risk	Hazard	Mitigation(s)

1.6. Results

L / cm	$\frac{1}{L}$ / cm^{-1}	$P \cdot 10^5$ / Pa
50.2	0.0199	1.10
47.8	0.0209	1.15
45.0	0.0214	1.20
25.0	0.0255	1.40
32.0	0.0280	1.55
39.0	0.0352	1.93



1.7. Analysis

$$P = 52.816\left(\frac{1}{L}\right) + 0.058$$

1.8. Uncertainties

It was rather difficult to get the pump to pump air into the capillary tube, which could allude to systematic errors.

1.9. Evaluation

The constant in the equation 0.058 is relatively small, which reflects random error and shows that the experimental data is relatively accurate

2. Practical 2: Charles' Law

2.1. Aim

To Investigate Charles' Law

2.2. Introduction

Given measuring the cross sectional area of the tube is rather difficult; the area stays constant throughout, therefore the length of the fluid in the tube is proportional to the volume of the fluid in the capillary tube.

$$\therefore L \propto V$$

2.3. Variables

- **Independent:** Temperature
- **Dependent:** Length (Volume)
- **Control:** Mass, Pressure

2.4. Method

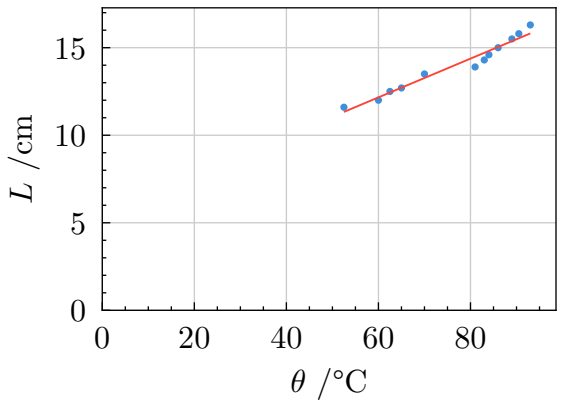
1. Set up the apparatus as shown in the diagram, ensuring the hot water in the beaker or measuring cylinder fully submerges the air sample.
2. Stir the water well using the thermometer, then read and record the initial temperature (θ_i /°C) and the length of the air sample (L /cm).
3. add ice and stir well, taking measurements every 5 °C down to room temperature.
4. plot a graph of θ /°C (x-axis) against L /cm (y-axis), drawing a best-fit line.
5. deduce the point of absolute zero, θ_0, from the graph by extrapolating the line to the x-intercept (where $L = 0$).

2.5. Risk Assessment

Hazard	Risk	Mitigation(s)
Boiling water	Spills and Burns skin	Handle boiling water with care, taking caution not to cause any spills.

2.6. Results

θ /°C	L /cm
93.0	16.3
90.5	15.8
89.0	15.5
86.0	15.0
84.0	14.6
83.0	14.3
81.0	13.9
70.0	13.5
65.0	12.7
62.5	12.5
60.0	12.0
52.5	11.6



$$L = 0.111\theta + 5.52$$

2.7. Analysis

Setting $L = 0$ gives $\theta_0 = -49.7\,^\circ\text{C}$, which is far from the true value of absolute zero temperature, since it is an extrapolation of data. Calculating a $\Delta\%$age:

$$\begin{aligned}\Delta\% &= \frac{-273+49.7}{-273} \times 100 \\ &= 81.8\% > 20\%\end{aligned}$$

$\therefore$ results are not accurate.

2.8. Uncertainties

2.9. Evaluation

Large errors, due to:

1. not being able to fully submerge the packet of air fully in the water
2. steam from the water making it difficult to read off of the scales

3. References

Boyle's and Charles' Laws *Accessed: 25/09/2026 @ 18:58*